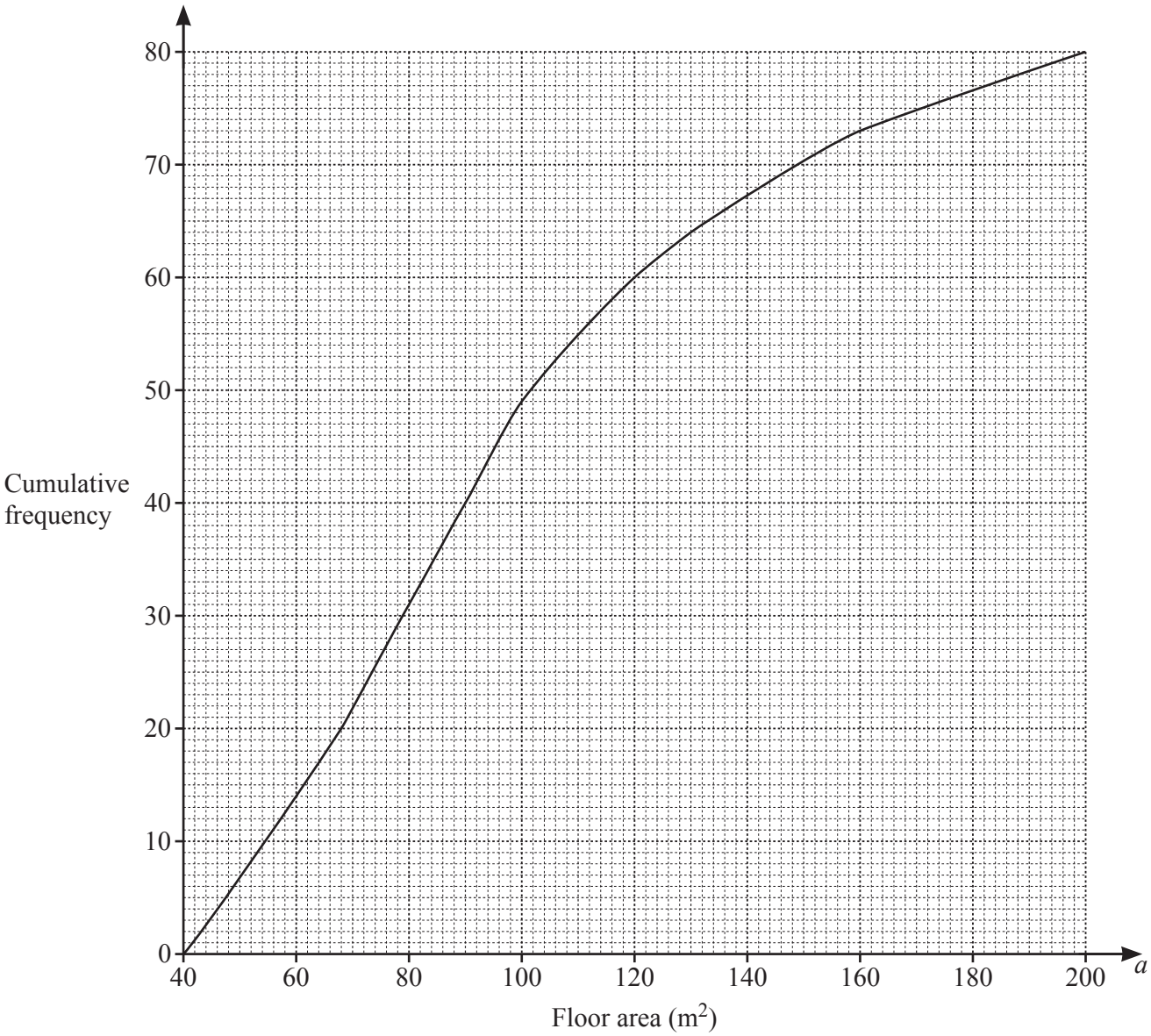
2 **(a)** The cumulative frequency diagram shows information about the floor area, $a\,\text{m}^2$, of each of 80 houses.



Use the diagram to find an estimate of

(i) the median,

..... m^2 [1]

(ii) the lower quartile,

..... m^2 [1]

(iii) the interquartile range,

..... m^2 [1]

(iv) the number of houses with a floor area greater than $120\,\text{m}^2$.

..... [2]

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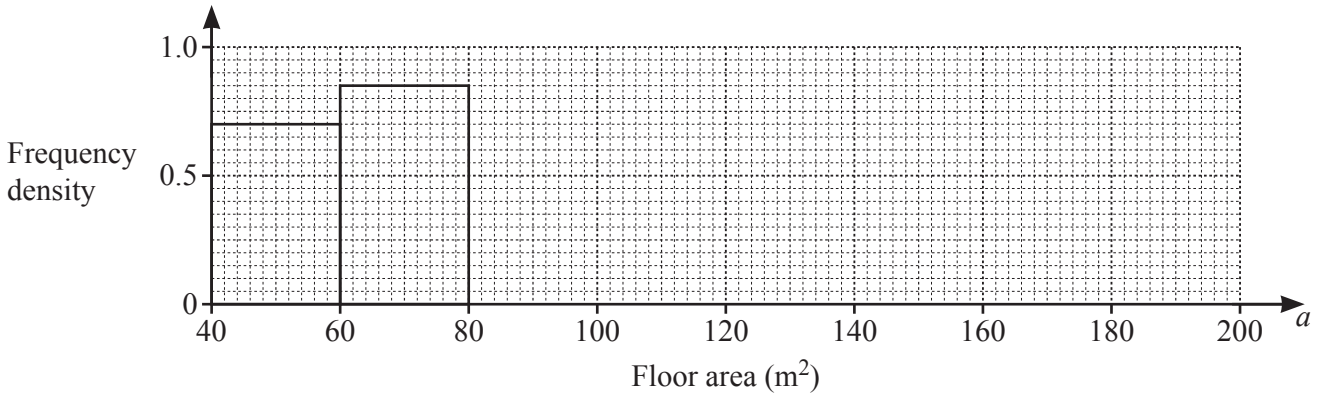
(b) The information about the 80 floor areas is shown in this frequency table.

Floor area (a m^2)	$40 < a \leqslant 60$	$60 < a \leqslant 80$	$80 < a \leqslant 100$	$100 < a \leqslant 130$	$130 < a \leqslant 160$	$160 < a \leqslant 200$
Frequency	14	17	18	15	9	7

(i) Calculate an estimate of the mean floor area.

..... m^2 [4]

(ii) Complete the histogram to show the information in the frequency table.



[4]

(iii) Two of the houses are picked at random.

Find the probability that one of the houses has a floor area greater than 130 m^2 and the other has a floor area 60 m^2 or less.

..... [3]